



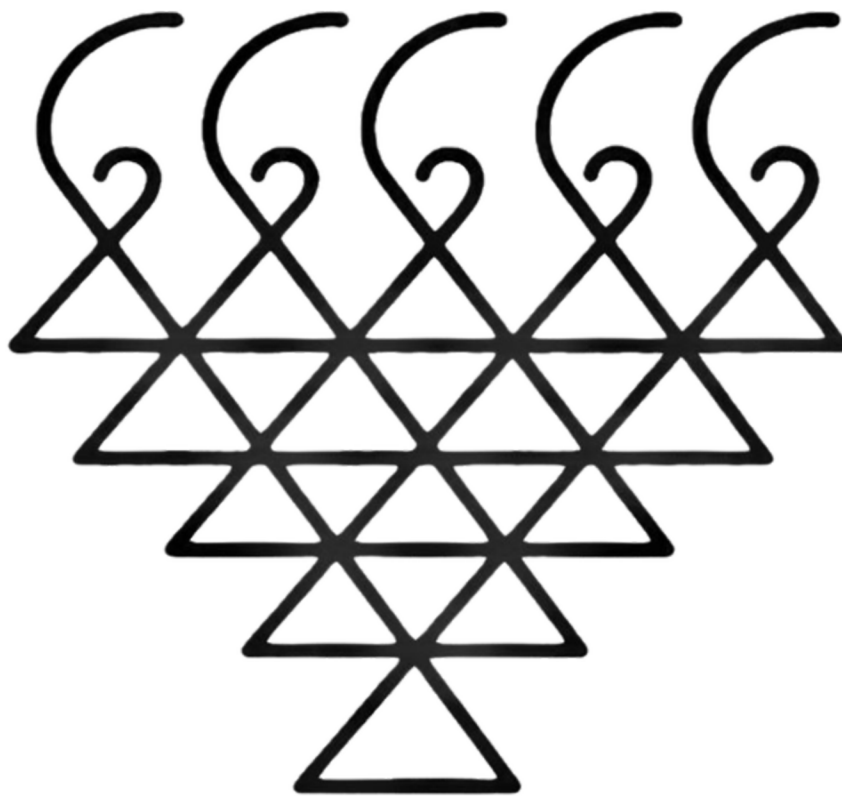
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ज्ञानेश्वरी

Advance Java



**“Every Student's Story,
Must be a Success Story!”**



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Topic 1. HTML Background

What is HTML?

- HTML (_____) is the standard language used to design the structure of web pages.
- It is not a _____ language, but a markup language (it uses tags to mark elements).
- A web browser (like Chrome, Firefox, Edge) reads the _____ file and renders it as a web page.

_____ means linking to other pages. _____ means using tags (<h1>, <p>, etc.) to format content.

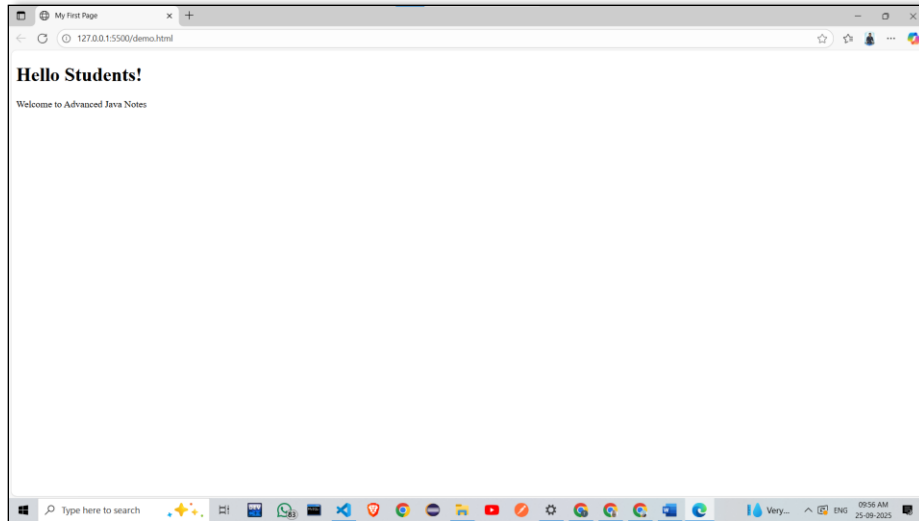
Features of HTML

1. Simple and Easy to Learn. Syntax is human-readable.
2. Platform _____. Works on any OS/browser.
3. Structuring Content. Defines headings, paragraphs, tables, images, forms, etc.
4. Supports Multimedia. Can embed images, audio, video.
5. Hyperlinking. Linking one page to another.

Basic Structure of an HTML Page

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>  <!-- Title shown in browser tab -->
</head>
<body>
  <h1>Hello Students!</h1>      <!-- Heading -->
  <p>Welcome to Advanced Java Notes</p>  <!-- Paragraph -->
</body>
</html>
```

Output :-



Explanation of Tags:

- **<!DOCTYPE html>** → Defines HTML5 version.
- **<html>** → _____.
- **<head>** → Metadata (title, CSS, scripts).
- **<title>** → _____.
- **<body>** → Visible content (text, images, links, forms).
- **<h1>** → Heading (h1 is biggest, h6 is smallest).
- **<p>** → Paragraph.

Common HTML Tags

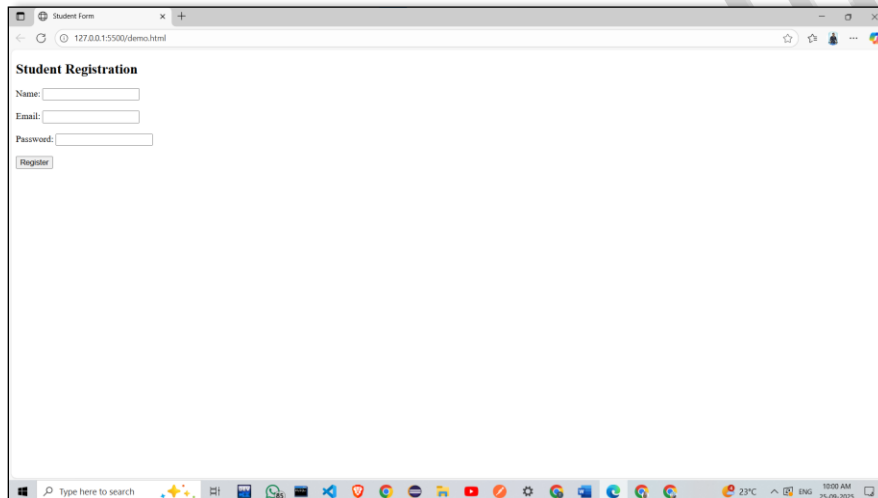
Tag	Purpose	Example
<h1> ... </h1>	Headings (H1–H6)	<h2>Chapter 1</h2>
<p> ... </p>	_____	<p>This is a paragraph</p>
	Hyperlink	Flipkart

<table>	Tables	<table><tr><td>Data</td></tr></table>
<form>	Forms	<form><input type="text"></form>

Example: Student Registration Form

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Form</title>
</head>
<body>
  <h2>Student Registration</h2>
  <form action="RegisterServlet" method="post">
    Name: <input type="text" name="name"><br><br>
    Email: <input type="email" name="email"><br><br>
    Password: <input type="password" name="password"><br><br>
    <input type="submit" value="Register">
  </form>
</body>
</html>
```

Output :-



The screenshot shows a web browser window titled "Student Form" with the address bar displaying "127.0.0.1:5500/demo.html". The page content includes a heading "Student Registration" followed by three input fields: "Name:", "Email:", and "Password:". Below these fields is a "Register" button. The browser's taskbar at the bottom shows various application icons and the system clock indicating 10:00 AM on 25-09-2023.

Here, when the student clicks Register, data will be sent to a Servlet for processing.

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Topic 2. Introduction to Web Applications

1. What is a Web Application?

A Web Application is a _____ that works using the _____ model and is accessed through a web browser such as Chrome, Firefox, Edge, etc.

Unlike desktop applications, a _____ application does not require installation on the user's device. The user only needs:

- A web browser
- An internet connection
- The URL of the application

Where does it run?

A _____ runs on a web server (Tomcat, Jetty, JBoss, Apache, Nginx, etc.). Users send requests to this server, and the server sends back a response.

2. Components of a Web Application

Every web application has two main parts, each serving a different purpose:

A. Frontend (Client-Side)

- This is the visible part of the application.
- Whatever users can see, click, type, and interact with is part of the frontend.
- Technologies used:
 - HTML :- _____ .
 - CSS :- _____ .
 - JavaScript :- Interactivity, form validations, animations, dynamic UI

Example:

When you open Flipkart, the homepage layout, product cards, search bar, add-to-cart button all are frontend elements.

B. Backend (Server-Side)

- This is the brains of the application everything happening behind the scenes.



- It handles:
 - Business logic (calculations, decisions, rules)
 - Database interactions (insert, update, delete, fetch)
 - Authentication and authorization
 - Processing user inputs
 - Generating dynamic responses
- Technologies used:
 - Java Servlets
 - JSP (Java Server Pages)
 - Spring / Spring Boot
 - Hibernate (for ORM)
 - Databases like MySQL, Oracle, MongoDB, PostgreSQL

Example:

When a user logs into Gmail:

- Backend verifies email + password.
- Fetches user data from the database.
- Returns inbox messages to the frontend.

Practical Understanding

When you click "Login", "Place Order", "Search", etc.:

- You trigger a request to the server.
- The server processes it and sends a response.

3. How a Web Application Works (Request–Response Cycle)

This flow explains exactly what happens when a user enters a _____ .

a. User enters a URL

- Example: <https://www.gmail.com>
- The browser prepares a _____ Request containing:
 - i. URL



- ii. Method (GET/POST)
- iii. Headers
- iv. Cookies (if any)
- v. Data (for POST requests)

b. Browser sends request to the _____ Server

- Server receives the request on a specific _____ (Tomcat uses port _____ by default).

c. Server-side components process the request

Depending on the application, the request is passed to:

- Servlet
- JSP
- Spring Controller (Spring MVC)

They perform:

- Business logic
- Form validations
- Processing user inputs
- Decision making

d. Database Interaction (if required)

If the request requires data:

- Server interacts with the _____
using JDBC/Hibernate/JPA to:
 - Insert new data
 - Retrieve existing data
 - Update records
 - Delete records

e. Response is generated

Server generates a _____ which may contain:

- HTML
- JSON
- XML
- Files (images, PDFs, videos)
- Error messages (404, 500, etc.)

f. **Browser receives and displays the _____**

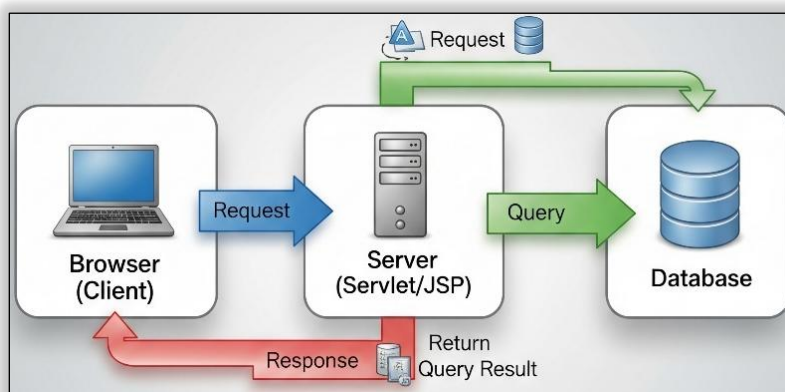
- Browser renders the _____
- User sees the final output on the screen

Example – Flipkart Product Search

1. User types "iPhone 15" in Flipkart search box
2. Browser sends request → /search?query=iPhone+15
3. Server (backend) receives the query
4. Backend searches the database for "iPhone 15"
5. Backend creates a list of matching products
6. Server sends HTML + product data
7. Browser displays the product list

This complete process takes less than 1 second, but a lot happens internally.

Flow Diagram :

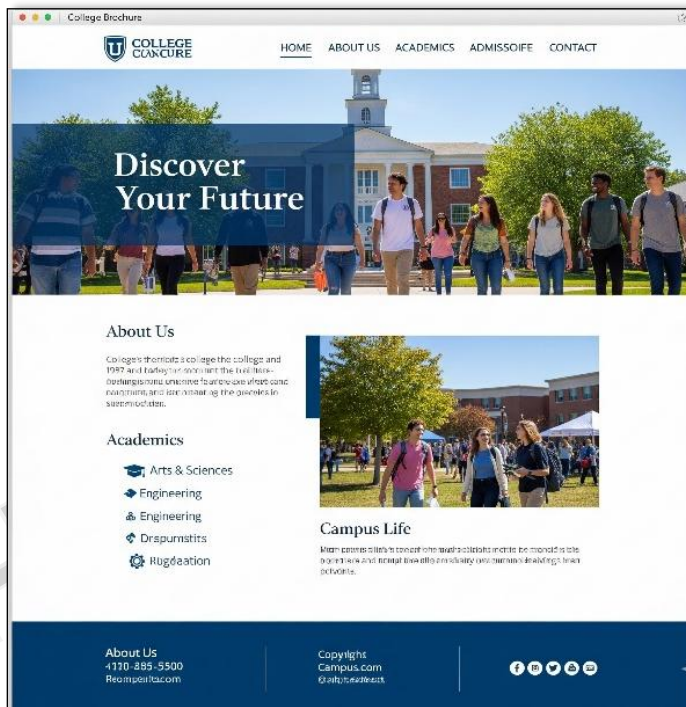


Types of Web Applications

1. _____ Web Applications

- Built with HTML, CSS (sometimes JavaScript).
- Content is _____ and does _____ change based on user input.
- Every user sees the _____ content.
- Examples:
 - College brochure site
 - Company's "About Us" page

Limitation: Cannot process logic or connect to databases.

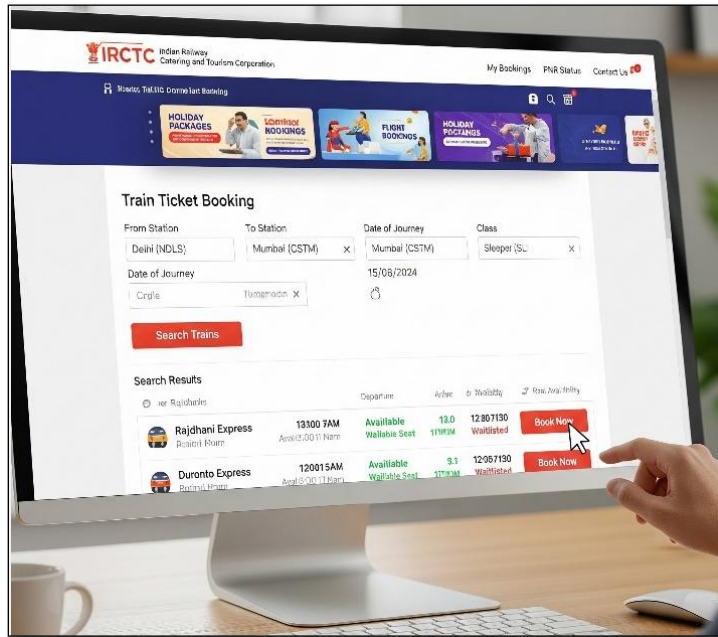


2. _____ Web Applications

- Built with Servlets, JSP, PHP, Python, Node.js, etc.
- Content is dynamic (changes for each user, request, or data).
- Interacts with databases to fetch/store information.
- Examples:
 - i. Flipkart (different products for each search)

- ii. Gmail (different inbox for each user)
- iii. IRCTC (different availability based on date/train)

Advantage: Personalized and interactive experience.



Advantages of Web Applications

1. Cross-_____ Compatibility

Since web apps run inside a browser, they work on any operating system or device without needing separate versions for each platform.

2. No Installation Required

Users can access the application directly through a _____, which removes the need for downloads and saves device storage.

3. Easy to Update

All updates are done on the _____, so every user automatically gets the latest version without any action on their side.

4. Scalable for Large User Bases

Web apps can be easily scaled by upgrading servers or adding more resources, allowing them to handle hundreds or thousands of users efficiently.

5. Database Integration

They support strong database connectivity, enabling features like user login, order management, payments, and personalized content.

Example: IRCTC (Train Ticket Booking System)

1. Frontend (HTML/JSP Interface)

On the IRCTC website, the user enters journey details such as the source station, destination, travel date, and passenger information. This part of the application is designed using HTML, CSS, JavaScript, and JSP to collect user input in a readable and interactive format.

2. Servlet / Backend Processing

When the user submits the form, the request goes to the backend where a Servlet or controller receives it. The backend reads the journey details, applies business logic, checks seat availability, validates data, and prepares the required instructions for the next steps.

3. _____

The backend interacts with the IRCTC database, which stores all information about trains, routes, schedules, seat availability, and existing bookings. The system fetches available trains matching the user's criteria and updates booking details whenever a ticket is confirmed.

4. Response Back to User

After processing the database results, the server sends a response back to the user's browser. This response contains the list of available trains, seat availability, fare, and booking confirmation status. The user then views these details on the screen and proceeds with booking.

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Topic 3. Layered Application Design

What is Layered Application Design?

Layered Application Design is a method of _____ by dividing it into separate layers, each responsible for a specific task. Instead of mixing all code together, the application is organized into logical sections such as _____, _____, and data handling that communicate with one another in an orderly way. This separation makes the application easier to maintain, more scalable, and highly reusable because changes in one layer do not directly impact the others. Overall, it provides a cleaner and more manageable architecture for building web applications.

The 5 Layers of Application Design (Correct Sequence)

1. _____ Layer (User Interface Layer)

The _____ Layer is the topmost layer of the application and is responsible for all user interactions. It _____ information to the user and collects inputs such as login details, registration data, or form submissions. This layer focuses only on _____ and does not contain _____. It is built using technologies like HTML, CSS, JavaScript, JSP, Angular, or React.

Example: A login page where the student enters their email and password.

2. _____ Layer

The _____ Layer acts as a mediator between the _____ Layer and the _____ Layer. It exposes a set of _____ that the UI can call, ensuring clean communication and reducing direct _____ on _____. This layer organizes application functionalities into reusable service methods.



Example: StudentService.registerStudent() is called by the _____, which then forwards the request to the Business Layer.

Technologies: Spring Services, REST APIs, Web Services.

3. _____ Layer (Business Logic Layer / Application Layer)

The _____ Layer contains the _____ and _____ logic of the application. It validates user requests, applies conditions, and processes operations before any database interaction occurs. This layer ensures that only correct and valid operations proceed further.

Example: Checking if the student meets the age criteria before allowing registration.

Technologies: Core Java, Spring, EJBs.

4. _____ Layer

The _____ Layer focuses on converting Java objects into database format and vice versa. It acts as a helper layer for writing _____ code or ORM logic. This layer uses _____ defined in the _____ Layer and contains the actual implementation behind them.

Example: Mapping a Student object to the students table through ORM.

Technologies: JDBC implementations, Hibernate, JPA, ORM frameworks.

5. _____ Layer

The _____ Layer defines all the interfaces related to _____. It contains method declarations for _____ operations, but not their _____. The _____ Layer provides the implementation of these interfaces. This separation ensures clean architecture, better testability, and easy maintenance.

Example: StudentDAO contains methods such as saveStudent(), getStudent(), updateStudent(), etc., which will later be implemented by persistence classes.

Example Flow: Student Registration (Using Layer Architecture)



1. _____ Layer

The user fills out the student registration form by entering details such as name, email, age, and contact information. The form submits this data to the application through the UI.

2. _____ Layer

The Service Layer receives the submitted data from the Presentation Layer. It acts as the communication bridge and forwards this data to the Business Layer for further processing. It ensures a clean separation between UI and business logic.

3. _____ Layer

The Business Layer applies all necessary rules on the submitted data. It verifies conditions like whether the email is unique, whether the student meets the minimum age requirement, and whether the data is valid. If everything is correct, the Business Layer passes the processed data back to the Service Layer to proceed with saving.

4. _____ Layer

The Persistence Layer uses the interfaces defined in the DAO Layer and contains the actual implementation for saving the student data. This layer converts the Student object into a format suitable for database operations (using JDBC or ORM tools like Hibernate/JPA).

5. _____ Layer

The DAO Layer contains the interfaces for all CRUD operations. The Persistence Layer implements these interfaces and executes the required SQL queries. The DAO methods are finally used to store the validated student data permanently into the database.

Advantages of Layered Design

- Each layer has its own role, so the structure is clear and organized.
- Easy to maintain because changes in one layer don't affect others.
- Business logic and services can be reused in different projects.
- Suitable for large and scalable applications.
- Easy to test since every layer can be checked independently.

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Topic 4. Application Stack

What is an Application Stack?

An Application Stack is the complete collection of _____, frameworks, and tools used to build and run a web application. It describes how different parts of the application like the user interface, backend logic, server, and database work together as one system. You can think of it as the “_____” that developers use to create a web app, where each tool has a role and all components combine to deliver the final application.

Components of an Application Stack

An Application Stack is made of several important components that work together to build a complete web application:

1. _____ (Client-Side)

This is the part of the application that the user directly sees and interacts with. It focuses on design, layout, and user experience. Technologies include HTML, CSS, JavaScript, JSP, Angular, React, etc.

Example: Login form, product list, shopping cart page.

2. _____ (Communication & Processing Layer)

This layer acts as a bridge between the Frontend and Backend. It handles request routing, data exchange, and sometimes light processing. Common technologies include APIs (REST APIs), Controllers, Filters, and Middleware. Example: Receiving user login input from frontend and passing it safely to backend services.

3. _____ (Server-Side)

This layer contains the application’s core logic. It processes user requests, applies business rules, and decides what data is needed.

Built using Java Servlets, Spring, Spring Boot, REST APIs, etc.

Example: Verifying login credentials, processing payments, generating reports.

4. Database



This is where all application data is stored permanently. It can be Relational (MySQL, Oracle, PostgreSQL) or NoSQL (MongoDB).

Example: Saving user accounts, orders, transactions, or product details.

Example of a Common Java Application Stack

- _____ : HTML, CSS, JavaScript, JSP
- _____ : Controllers, Filters, REST APIs (used for communication between frontend and backend)
- _____ : Java Servlets, Spring Framework
- _____ : MySQL, Oracle, PostgreSQL
- **Server/Container:** Apache Tomcat or GlassFish

Why Application Stack is Important?

An application stack plays a crucial role in building a well-structured web application. It ensures that the frontend, middle layer, backend, and database work together smoothly to deliver the final output to the user. By using a defined set of technologies, developers can follow a proper structure, build scalable systems, maintain security, and easily update or expand the application. It also allows different technologies like Java for backend, MySQL for database, and React for frontend to be integrated into one complete system.

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Topic 5. Introduction to Servlets

What is a Servlet?

A Servlet is a _____ used to create _____ meaning it can generate content based on user inputs, database values, or business logic. When a user sends a _____ from a browser such as Chrome or Firefox, the _____ receives that request on the server, processes it, and sends back an appropriate response like an HTML page or data result. _____ always run inside a special environment known as a _____ (such as Apache Tomcat or Jetty), which manages their lifecycle and handles communication between the client and the server.

Key Features of Servlets

1. Server-Side Technology

Servlets run entirely on the _____, not on the user's machine. This allows them to handle client _____, process business logic, and generate responses efficiently.

2. Dynamic Content Generation

Unlike static HTML pages, Servlets can create dynamic output. They generate responses based on user inputs, database values, or real-time calculations, making web applications interactive and personalized.

3. Platform Independent

Since Servlets are written in _____, they can run on any operating system or server that supports the Java platform, ensuring easy portability across environments.

4. Secure and Reliable

Servlets inherit _____ built-in security features like exception handling, garbage collection, and strong _____. This makes them more secure and less error-prone compared to many other server technologies.

5. Highly Efficient (Multithreading)

Servlets use _____, meaning one _____ instance can serve multiple users at the _____. This reduces memory usage, improves performance, and helps the application scale smoothly for many users.

Why Servlets?

Before Servlets were introduced, web applications commonly used CGI (_____) to handle requests. However, CGI had major performance limitations because it created _____



a new process for every single client _____. This made it extremely slow and resource-heavy, especially when many users accessed the application at the _____.

Servlets solve this problem by using multithreading instead of process creation. A single Servlet instance can handle multiple requests through separate threads, which makes request processing much faster, more efficient, and less memory-consuming. This improvement allows modern web applications to support thousands of users smoothly.

How a Servlet Works (Flow)

1. Client Sends a Request

The user performs an action in the browser such as submitting a login form and the browser sends an _____ request to the server.

2. Web Server Passes the Request to the Servlet Container

The web server receives the _____ and forwards it to the Servlet Container (like Tomcat), which is responsible for managing _____.

3. Servlet Container Creates or Reuses a Servlet Object

If the _____ is not loaded, the _____ creates an instance of it; otherwise, it uses the already-loaded instance to handle the _____.

4. Servlet Processes the Request

The _____ reads the data sent by the user, executes business logic, interacts with the database if needed, and prepares the output.

Example: Checking username and password in the database.

5. Servlet Sends Back a Response

It generates the final output (like an HTML page, JSON, or message) and sends it back to the _____.

6. Browser Displays the Result

The _____ is returned to the user's browser, which displays the result such as "Login Successful" or "Invalid Credentials".

Advantages of Servlets

1. Better Performance

_____ use multithreading instead of creating a new process for every _____. This makes them faster, lighter, and more efficient than older technologies like _____.

2. Reusable and Easy to Maintain

Since Servlets are written as _____, the code can be reused across multiple parts of the application. Updates or modifications can also be done easily without affecting the whole system.



3. Robust and Secure

_____ inherit Java's strong security features such as exception handling, type safety, and _____. This makes them reliable and secure for web applications.

4. Cross-Platform Compatibility

Any machine that has a _____ can run Servlets, making them _____ - _____ and easy to deploy on different _____.

5. Highly Extensible

Servlets can easily integrate with _____ frameworks like Spring, Hibernate, and JSP, allowing developers to build more powerful and feature-rich applications.

Example : Understanding Servlets Through a Restaurant Example

Think of a web application as a restaurant. When a customer visits the restaurant, there is a clear flow of communication this is similar to how a Servlet works between the user and the server.

- **Customer (Browser):**

The customer walks into the restaurant and places an order. In web terms, this is the browser sending a request to the server for example, submitting a login form.

- **Waiter (Servlet):**

The waiter listens to the customer, writes down the order, and takes it to the kitchen. This is exactly how a Servlet works. It receives the request, understands what the user wants, and forwards it for processing.

- **Kitchen (Server + Business Logic + Database):**

The kitchen prepares the order based on the customer's request. Similarly, the server processes the request, performs business logic, and may interact with the database (like checking login details or retrieving product information).

- **Waiter Returns (Servlet Response):**

Once the food is ready, the waiter brings it back to the customer. In the same way, the Servlet sends the processed result back to the user's browser—such as a success message or a web page.

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Topic 6. Servlet Structure

1. Servlet Interface

The _____ interface (from javax.servlet.Servlet) is the _____ interface for all types of Servlets in Java. Every servlet must either implement this interface directly or indirectly through other classes like GenericServlet or HttpServlet. The interface defines _____ essential methods that control the _____ and behavior.

1. **init(ServletConfig config)**

This method is called only _____ when the servlet is first created. It is used for initialization tasks such as reading configuration parameters.

2. **service(ServletRequest req, ServletResponse res)**

This is the core method of the servlet. It is called for every _____ and is responsible for processing the request and generating the response.

3. **destroy()**

Called when the servlet is about to be removed from _____. It is used to close resources, release connections, or perform cleanup operations.

4. **getServletConfig()**

Returns the _____ object containing initialization and configuration information about the servlet.

5. **getServletInfo()**

Returns a brief description of the servlet, such as author information or version details.

Drawback:

When a class directly implements the _____ interface, it must provide implementations for all _____ methods, even if some of them are not needed. This makes the approach less convenient, which is why developers usually extend _____ or _____ instead.

```
public class MyServlet implements Servlet {
    public void init(ServletConfig c) {}
    public void service(ServletRequest req, ServletResponse res) {}
    public void destroy() {}
    public ServletConfig getServletConfig() { return null; }
    public String getServletInfo() { return "info"; }
}
```



2. GenericServlet (Abstract Class)

GenericServlet is an abstract class provided in the _____ package, designed to simplify servlet development. It already implements the _____ interface, which means you don't have to write all _____ methods yourself. Instead, it provides default (empty) implementations for most lifecycle methods and leaves only the _____ method as abstract. Because of this, when you extend _____, you only need to override the _____ method to handle client requests. Also, since it is not tied to any specific protocol, _____ can be used for various communication protocols beyond just _____, making it more flexible for developing generic _____ components.

```
public class MyServlet extends GenericServlet {  
    public void service(ServletRequest req, ServletResponse res) {  
        res.getWriter().println("Hello from GenericServlet");  
    }  
}
```

3. HttpServlet (Preferred Approach)

_____ is a class from the _____ package and is the most commonly used way to create _____ in modern web applications. It is a subclass of _____ but is specifically designed to work with the _____ protocol, which is the foundation of all web communication. Instead of using a single _____ method, HttpServlet provides specialized methods to handle different types of _____ requests. For example, _____ is used for handling _____ requests such as opening a webpage, while doPost() is used for _____ requests like form submissions. It also includes methods like _____ and doDelete() for advanced operations. Because it simplifies handling of each _____ request type, _____ is the preferred and widely used approach for building dynamic Java web applications.

```
public class MyServlet extends HttpServlet {  
    public void doGet(HttpServletRequest req, HttpServletResponse res) {  
        res.getWriter().println("Hello from HttpServlet");  
    }  
}
```

Comparison

Feature	Servlet Interface	GenericServlet	HttpServlet (Best)
Package	javax.servlet	javax.servlet	javax.servlet.http
Methods required	All 5	Only service()	doGet(), doPost(), etc.
Protocol support	Any	Any	HTTP only
Usage	Rare	Less common	Most common

Hierarchy Diagram :



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Topic 7. Servlet Life Cycle

Definition :

The Servlet Life Cycle describes the complete journey of a servlet from the moment it is created, through handling multiple client requests, until it is finally removed from memory. All these steps are controlled by the _____ (such as Tomcat, Jetty, or GlassFish), which manages the servlet's creation, execution, and cleanup.

The life cycle is mainly defined by three key methods:

_____ for initialization, _____ for processing client requests, and _____ for cleanup before the servlet is unloaded.

Lifecycle Stages:

1. Class Loading

Class loading is the first step in the servlet life cycle. In this stage, the Servlet Container loads the servlet's _____ file into memory. This loading can happen in two situations:

- **When the first client request arrives** : the container loads the servlet only when it is needed.
- **During server startup** : if the servlet is configured with the <load-on-startup> option in web.xml, the container loads it as soon as the server starts.

This step prepares the servlet so the container can create its object and begin execution.

2. Object Creation

After loading the servlet class, the Servlet Container creates a _____ instance (object) of that servlet. This object is created only _____ during the servlet's entire life cycle. The same servlet instance is then shared for handling all incoming client requests.



Instead of creating new objects every time, the container uses multithreading, where each client request is processed by a separate thread. This approach greatly improves performance and reduces memory usage.

3. Initialization → init()

Once the servlet object is created, the Servlet Container calls the _____ method only one time. This method is used to perform all one-time initialization tasks before the servlet starts handling client requests. Typical activities done inside init() include:

- Establishing database connections
- Loading configuration or property files
- Initializing resources required throughout the servlet's lifetime

After _____ completes, the servlet is fully ready to process incoming requests.

```
public void init() throws ServletException {  
    System.out.println("Servlet Initialized...");  
}
```

4. Request Handling :- service()

The _____ method is executed every time a client sends a request to the _____. Unlike init(), which runs only once, the service() method is called repeatedly—once for every request. Its main job is to read the incoming request, apply the required logic, and generate the correct response for the client.

In the case of _____, the _____ method automatically determines the type of _____ request and then calls the appropriate method internally:

- Calls doGet() for _____ requests
- Calls doPost() for _____ requests



This mechanism ensures that the servlet handles each request correctly based on the HTTP method used.

```
public void service(ServletRequest req, ServletResponse res)
    throws ServletException, IOException {
    res.setContentType("text/html");
    PrintWriter out = res.getWriter();
    out.println("<h1>Hello from Service()</h1>");
}
```

5. Destruction :- destroy()

The _____ method is called only once, just before the servlet is removed from memory usually when the server is shutting down or the application is being stopped. This step allows the servlet to perform all necessary _____ tasks to ensure that no resources are left open or unused. Typical cleanup activities include:

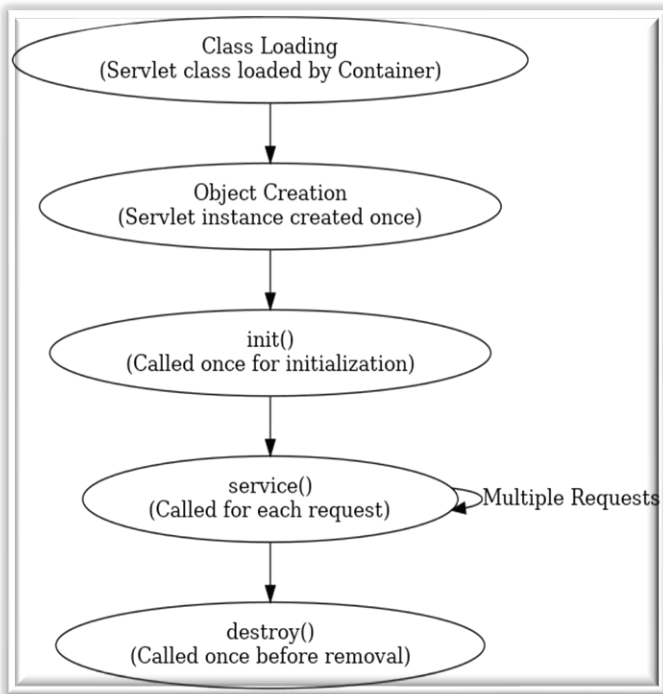
- Closing active database connections
- Releasing files, network links, or memory resources
- Saving any final data if required

Once _____ completes, the _____ is fully removed from memory, ending its life cycle.

```
public void destroy() {
    System.out.println("Servlet Destroyed...");
}
```



Servlet Life Cycle :





Topic 8. Inter-Servlet Communication & Sessions

What is Inter-Servlet Communication?

Inter-Servlet Communication means one _____ interacting or communicating with another _____ inside the same web application. This is important because in real projects, different servlets handle different tasks, and they often need to share data or pass control between each other.

For example, one servlet may handle login validation, while another servlet displays the user dashboard. To make this flow work smoothly, servlets must communicate.

Why is it needed?

In large web applications, different tasks are handled by different _____. No single servlet performs all the operations. Each _____ is responsible for a specific part of the workflow.

Because of this division of work, servlets often need to interact with each other to complete a process smoothly.

Example

- One servlet handles login validation
- Another servlet loads the user dashboard or profile

To move from login to the dashboard, these servlets must communicate and share data. This communication ensures proper flow, modular design, and better maintainability of the application.



How Servlets Communicate

Servlets communicate with each other using mechanisms that allow one servlet to pass control, share data, or redirect the user to another resource. These mechanisms help maintain a smooth flow between different parts of a web application.

Main Mechanisms of Inter-Servlet Communication

1. _____

Used when one servlet wants to _____ the same request to another servlet or include its output within the current response.

- _____() : Transfers the same request internally to another servlet.
- _____() : Inserts the output of another servlet into the current response.

2. _____

Sends a new request from the browser to another servlet or even an external website. The browser is aware of the redirection because the URL changes.

1. _____

_____ is used when one servlet needs to pass control to another _____ or _____ on the _____. It allows multiple components to work together in a _____ request flow.

Two Important Methods

I. _____(request, _____)

This method transfers the same request from one _____ to another internally.

- The transfer happens on the _____ side.
- The browser is *not* aware of it - _____ remains unchanged.
- The second servlet/JSP continues processing the same request and response object.
- Common Use Case:

After a successful login, LoginServlet forwards the request to WelcomeServlet to display the dashboard.



II. _____(request, _____)

The _____() method is used when one servlet wants to insert the output of another servlet or JSP into its own response.

- The included _____ output becomes part of the same final page.
- Both components contribute to one combined _____.
- The original _____ continues executing after the include.
- Common Use Case

Adding a common header, footer, sidebar, or menu to multiple JSP pages.

2. _____()

_____() is used when a servlet wants to _____ the client's browser to another resource. Unlike _____(), this method creates a new _____, and the browser is fully aware of the _____ because the _____ in the address bar changes.

The destination can be anywhere either within the same web application or even an external website like google.com.

Example

After _____, the servlet redirects the user to login.html:

```
response.sendRedirect("login.html");
```

Difference Table

Feature	forward()	sendRedirect()
Request type	Same request	_____
URL change?	No	Yes
Server/External?	Only within server	Both internal/external
Speed	_____	Slower (extra trip)



What are Sessions?

A _____ is a mechanism used to store user-specific information across multiple requests in a web application. Since _____ is a stateless protocol, the server does not automatically remember who the user is after each request. Sessions solve this problem by maintaining continuity between _____.

They are commonly used to preserve data such as login status, shopping cart items, and user preferences while the user _____ through the application.

Common Session Management Techniques

1. _____
2. _____
3. _____

1. HttpSession

- The most widely used technique for managing sessions in Java _____ applications.
- Stores user-specific data directly on the _____, which makes it more secure because sensitive information is not exposed to the browser.
- Each user gets a unique _____, usually stored in a cookie or passed through URL rewriting.
- Suitable for login systems, dashboards, shopping carts, and multi-step user processes.

2. Cookies

- Small text files saved in the user's _____ by the server.
- Can store simple information like _____, _____, or _____.
- Cookies are limited in size (around _____) and may have an expiry time to control how long the data is stored.



- They are commonly used when the website needs to remember the user even after the browser is closed (e.g., “Remember Me” feature).
- Less secure than server-side storage, so sensitive data should not be stored directly in _____.

3. URL Rewriting

- Used when cookies are disabled or blocked by the _____.
- The session ID is attached to the _____ (e.g., dashboard.jsp;jsessionid=12345).
- Ensures that the server can still track the user session without relying on cookies.
- Should be used carefully because _____ can be visible, copied, and shared, which may expose the session ID.

Why Sessions Are Needed?

In web applications, sessions are required because _____ is a stateless protocol. This means every time the browser sends a _____ to the _____, the server treats it as a completely new request. It does not remember who the user is or what they did previously. Because of this, important features like login, shopping cart, user dashboard, and multi-step forms cannot work unless we store user information somewhere.

To solve this problem, _____ help the server remember the user across multiple requests. A session stores _____ data temporarily during the entire visit.

Where sessions are needed?

- When maintaining login information
- When storing shopping cart items
- When showing personalized pages (dashboard/profile)
- When handling multi-page forms
- When saving temporary preferences or actions



3. HttpSession

_____ is a built-in Java mechanism that allows the server to maintain _____ data across multiple requests. When a user first visits the application, the server creates a _____ session object for that user. Any data stored in this session is kept on the server, making it secure and _____ from the user's browser.

The session continues to exist as long as the user is _____. Once the user logs out or becomes inactive for a certain period (session timeout), the _____ automatically removes the session. _____ is commonly used to store information such as username, user role, cart items, or temporary data required during the user's visit.

```
HttpSession session = req.getSession();  
session.setAttribute("username", "Aadesh");  
String name = (String) session.getAttribute("username");
```

4. Cookies

Cookies are small pieces of _____ that the server stores on the client's browser. Every time the user sends a new _____, these cookies are automatically sent back to the _____, allowing the server to recognize the user or remember certain details. _____ are commonly used for lightweight, non-sensitive data such as remembering a username, saving user preferences, or maintaining "_____ " functionality on login pages. Since cookies are stored on the client side, they have size limits and should not contain sensitive information unless properly-encrypted.

```
Cookie ck = new Cookie("username", "Aadesh");  
res.addCookie(ck);
```

5. URL Rewriting

URL Rewriting is a technique where the _____ information is added directly to the URL so the server can identify the user without relying on _____. It is mainly used when cookies are disabled or not supported by the browser. In this method, data such as _____ or user information is passed as part of the query string in the URL.



Example:

<http://localhost:8080/app/WelcomeServlet?username=Aadesh>

In this example, the username is attached to the URL, and the servlet can read it to maintain the user's session or continue the flow of the application.

Comparison

Technique	Stored At	Security	Usage Example
HttpSession	_____	High	Shopping cart
Cookies	Client	_____	Remember login
URL Rewriting	URL	Low	_____

=====



Topic 9. Developing a Full Stack Application

1. Role of Frontend, Backend, Service Layer, Persistence Layer & DAO Layer

1. Frontend (_____ Layer)

The Frontend, or _____ Layer, is the visible part of the application that users interact with. It is built using technologies like HTML, CSS, JavaScript, Bootstrap, React, Angular, and other UI frameworks. This layer collects user input such as login details, search queries, or form submissions and displays the processed results sent by the backend.

Example:

Flipkart's homepage displaying product images, categories, prices, and offers is handled entirely by the frontend.

2. _____ Layer (Servlet Layer)

The _____ Layer acts as a bridge between the Frontend and the Main Business Logic. In Java applications, this layer is usually implemented using _____, which receive the incoming requests, read user data, and forward the request to the Business Logic Layer for processing. After receiving the processed result, the Service Layer sends the response back to the suitable JSP/HTML page.

Example:

When a user submits a login form, the Servlet reads the username/password, calls the Business Logic Layer to validate them, and then forwards the result to a JSP page for display.

3. _____ Layer (Core Processing Layer)

The _____ Layer performs all internal processing and decision-making. It ensures that the operations requested by the user follow the application's rules.



This is where _____, _____, _____, and _____ are-applied.

Responsibilities:

- Validating login credentials
- Calculating product prices and discounts
- Checking stock availability
- Applying age or eligibility rules

This layer does not deal directly with the _____ or _____ it only focuses on logic.

Example:

When a user adds an item to the cart, this layer calculates the updated total, discount, delivery charges, and verifies availability.

4. _____ Layer (Implementation Layer)

The _____ Layer handles the actual interaction with the _____. It converts _____ into database records and vice versa. This layer contains the _____ classes that perform database operations using _____, Hibernate, or JPA.

It uses the _____ defined in the _____ Layer and provides the real implementation behind them.

Example:

Saving a new user, inserting order details, updating product quantities, or mapping Java objects to database tables.

5. _____ Layer (DAO Layer – Interface Layer)

The _____ Layer defines standardized interfaces for database operations. It contains method _____ for _____ operations (Create, Read, Update, Delete) but not their implementations. This ensures _____ coupling and clean architecture. The Persistence Layer implements these interfaces.



Example:

Interfaces like UserDao, ProductDao, OrderDao with methods such as saveUser(), getUserById(), updateOrder(), etc. The Persistence Layer provides the actual logic for these methods.

6. Database (Persistence/Data Storage Layer)

The Database is responsible for permanent storage of all important application data users, orders, payments, bookings, product details, and transactions. Java applications connect to these databases using JDBC, Hibernate, or JPA.

Common Databases:

- Relational: _____, _____, PostgreSQL
- NoSQL: MongoDB

Example:

Storing login credentials, booking history, payment status, or updating product inventory.

2. Using Servlets as Backend

In Java web applications, _____ **act as the** _____, meaning they handle incoming requests, process the required logic, interact with the _____, and then send the final response back to the user interface. They work as the _____ **manager** between the frontend and the database.

Workflow of a Servlet as Backend

1. Receives the request coming from the frontend, such as an _____ form submission or an AJAX call.
2. Reads the user input, like username, password, search query, or form data.
3. Applies business logic, including validation checks, calculations, and decision-making.
4. Connects to the database using _____ or ORM tools to fetch, insert, update, or delete data.



5. Forwards the processed result to a _____ or HTML page for displaying the output to the user.

3. Flow of Request/Response

The request–response flow explains how a user’s action on the browser travels through the server, gets processed, and returns as a final output. This is the core working mechanism of every web application using _____.

Step by Step Flow

1. User Action

The user performs an action in the browser such as clicking *Submit*, *Login*, or *Search*. This triggers the start of the request.

2. Request Sent

The browser prepares an _____ request and sends it to the web server. This request contains all the data entered by the user.

3. Servlet Receives

The server _____ the request to the appropriate Servlet, which begins processing it.

4. Business Logic Execution

Inside the Servlet, all _____, _____, and _____ take place based on the user input and application rules.

5. Database Interaction

If required, the Servlet communicates with the _____ fetching or storing data using _____ or ORM frameworks.

6. Response Prepared

After processing, the Servlet prepares the final response.

This may be forwarded to a _____ page, or the Servlet may directly send _____ data in case of AJAX calls.



7. Browser Display

The browser receives the final response and displays it to the user—for example, a dashboard page, a booking confirmation, or an error message.

1. Example: Online Shopping (Mini Case Study)

- _____: User selects product on website.
- _____ (Servlet):
 - Reads selected product.
 - Validates stock availability.
 - Calculates total with discounts.
- _____: Updates cart, stores order.
- _____: Shows updated shopping cart page with “Order Successful”.

2. Key Advantages of Full Stack Approach

A full stack approach provides a structured way to build complete web applications by separating the system into layers such as UI, business logic, and database. This separation ensures that each part of the application works independently and can be improved without disturbing the others.

Major Advantages

- **Clear Separation of Concerns**

The frontend, backend logic, and database are _____, making the system organized and easier to manage.

- **High Maintainability and Scalability**

Developers can _____ or _____ one layer such as UI or backend without affecting the entire application, helping the system grow smoothly.

- **Reusable Code**

Backend components like _____ can be reused by multiple JSPs or pages, reducing duplication and speeding up development.

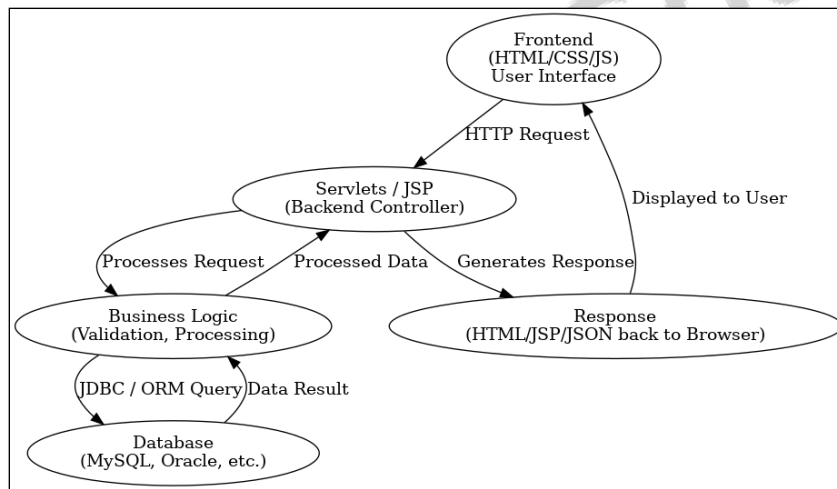
- **Better Security**

The backend handles data _____, _____ management, and access control, keeping the application more secure.

- **Suitable for Real-World Applications**

This approach supports complex systems such as banking portals, e-commerce platforms, and ticket booking systems where multiple operations must work together seamlessly.

3. Flow Diagram :





Topic 10. Need for JSP and Basics

1. Limitations of Servlets (Why JSP was introduced?)

Servlets are excellent for handling backend processing, but they are not suitable for building the _____. When developers tried to generate web pages using Servlets, several issues appeared. The biggest problem was the mixing of Java code and _____ inside the same file. Writing HTML using `out.println()` statements made the code messy, difficult to read, and almost impossible for designers to work with. Because of this, even small UI changes required modifying _____, _____ the servlet, and _____ the application a very time-consuming process.

Another drawback was the lack of separation of concerns. Both business logic and presentation logic were placed together in _____, making large projects hard to maintain and scale. As applications grew bigger, managing UI inside servlets became extremely inefficient and unorganized.

Conclusion

_____ are powerful for processing requests and applying business logic, but they are not ideal for creating web pages or handling _____.

_____ was introduced specifically to solve these UI and maintainability problems.

2. Introduction to JSP (Java Server Pages)

JSP, or Java Server Pages, is a server-side technology used to create _____ in Java-based applications. It closely resembles _____, which makes it easy for designers to work with, but it also supports embedding Java code directly within the page. This helps developers control the _____ behaviour of the UI while keeping the layout simple.

When a JSP page is executed, the server (such as Tomcat) automatically converts the _____ into a _____, compiles it, and then runs it. This means _____ combines the simplicity of HTML with the power of Java processing behind the scenes.

Where JSP Is Commonly Used

1. Displaying data and UI elements to the user
2. Creating and handling input forms



3. Showing outputs or results returned by backend logic
4. JSP plays an important role in separating presentation from business logic, making Java web applications cleaner and easier to maintain.

3. Advantages of JSP over Servlets:

JSP provides several improvements over Servlets, especially in terms of presentation and UI development. Unlike Servlets, JSP keeps the HTML clean and readable while allowing Java code only where it is actually needed. This separation makes the UI easier to design and maintain.

One of the strongest advantages is _____. JSP allows including external files like `header.jsp` or `footer.jsp`, which helps maintain consistency across multiple pages. JSP also speeds up development because UI changes do not require recompiling or redeploying the entire servlet. You can simply edit the JSP file, and the server automatically updates it.

JSP comes with several built-in objects such as _____, _____, _____, and _____, which are ready to use without extra code. It also supports tag libraries like JSTL and custom tags, reducing the need for Java _____ in the page.

Finally, JSP fits perfectly into the MVC architecture JSP handles the _____, Servlets act as the _____, and the database layer serves as the _____. This makes Java web applications cleaner, modular, and easier to maintain.

4. Basic JSP Tags

I. _____ Tag → `<% ... %>`

- For writing Java code inside JSP.

```
<% int num = 10; out.println("Number: " + num); %>
```

II. _____ Tag → `<%= ... %>`

- Directly prints value to response.

```
<h2>Hello <%= request.getParameter("user") %></h2>
```

III. _____ Tag → `<%! ... %>`

- Declares variables/methods at page level.

```
<%! int count = 0; %>
```

IV. _____ Tag → `<%@ ... %>`

- Gives instructions to JSP engine.



▪ **Examples:**

`<%@page import="java.util.*" %>`

`<%@include file="header.jsp" %>`

- V. _____ **Tag** → `<%-- comment --%>`
- For JSP-specific comments (not sent to client).

`<%-- This is server-side comment --%>`

5. Example JSP Flows

When a user accesses a JSP page, the request goes through a simple but important process. First, the user sends a request to a page such as *index.jsp*. The _____ receives this request and the JSP page is automatically converted into a _____ by the JSP container. Once converted, the generated servlet is compiled and executed. During execution, it processes any dynamic content like reading data, displaying user-specific information, or interacting with backend logic.

After the servlet completes its work, it produces the final _____ output combined with dynamic data, and this _____ is sent back to the user's browser. The user then sees the fully generated web page.

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Topic 11. JSP as Part of Full Stack Application

JSP plays an important role in the presentation layer of a Java Full Stack application. While Servlets handle backend processing and business logic, JSP focuses on displaying results, collecting input, and creating user-friendly web pages. Together, JSP and Servlets form the foundation of Java-based web applications.

JSP with Servlets

JSP works closely with _____ to build complete Java web applications by clearly separating the user interface from the backend processing. In this structure, Servlets function as the controllers that manage incoming _____, execute business logic, interact with databases, and prepare the data needed by the application. JSP, on the other hand, acts as the _____ layer, focusing only on presenting this processed data in a clean and user-friendly _____ format.

The general flow is simple: a JSP page collects user input such as login details, registration data, or search queries. This input is then sent to a Servlet, which processes the information, applies validations, communicates with backend services or the database, and generates the required result.

Once processing is complete, the Servlet forwards the data back to a JSP using a `RequestDispatcher`, allowing the page to display the output.

Flow Between JSP and Servlet:

1. JSP collects _____ from the user.
2. Servlet receives the _____ and applies business logic.
3. Servlet interacts with the database if required.
4. Servlet _____ processed data to _____.
5. JSP displays the final result.

This combination is powerful because JSP keeps the UI simple and easy to modify, while Servlets maintain well-organized backend logic. This separation prevents the mixing of HTML and Java code, resulting in cleaner, modular, and more maintainable applications suitable for real-world development.

MVC Pattern (Model – View - Controller)

The MVC pattern is a software design architecture that separates an application into three interconnected components - _____, _____, and _____ to separate concerns and make



development easier to manage. The _____ represents the application data and business rules; it knows how to retrieve, store, and update information (for example, using DAO and persistence code). The _____ is the presentation layer that displays data to the user (for example, JSP pages or React components). The _____ acts as the intermediary: it receives user input, calls the appropriate business logic in the Model, and decides which View should render the result.

This separation improves maintainability, enables parallel development (UI designers work on Views while backend developers work on Models/Controllers), and makes testing simpler because each part can be tested independently. It also supports reusability: the same Model can serve multiple Views, and Views can be reused with different Controllers.

Typical MVC Flow :

1. User interacts with the UI (View). for example, submits a login form.
2. The Controller receives the request and extracts input data.
3. Controller calls the Model (business/service layer) to process the request (validate credentials, fetch/update data).
4. Model interacts with the database (via DAO/Persistence) and returns results to the Controller.
5. Controller selects the appropriate View and forwards the Model data to it.
6. View renders the final response (HTML/JSON) and sends it to the user.

Login Example (MVC)

This example demonstrates a simple login flow: the user types credentials into a JSP form (View). The form posts to a LoginServlet (Controller) which calls an AuthService (Business Layer). The service uses UserDao (Data Access Layer) to check credentials in the database (Model / Persistence). On success the servlet creates a session and forwards/redirects to welcome.jsp; on failure it forwards back to login.jsp with an error.

Sequence

1. User opens login.jsp and submits username & password.
2. LoginServlet.doPost() receives the request.
3. LoginServlet calls AuthService.authenticate(username, password).
4. AuthService calls UserDao.getByUsername(username) to fetch user record.
5. If password matches (use hashed compare in real apps) → success; else → failure.
6. On success: set HttpSession attributes and forward to welcome.jsp (or sendRedirect). On failure: forward back to login.jsp with error.